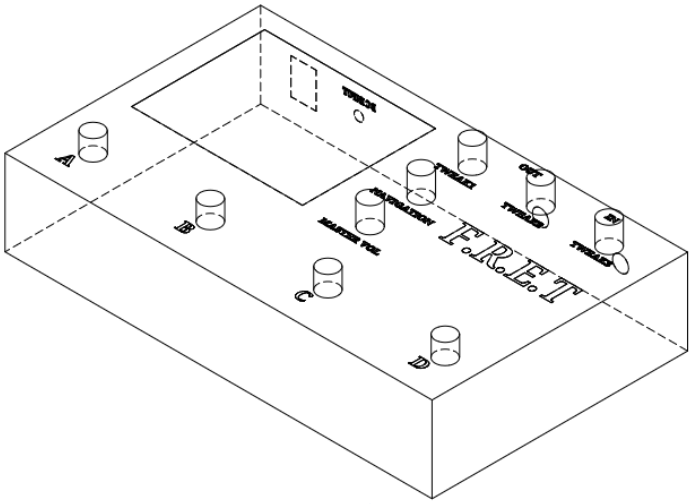


F.R.E.T. - Flexible Real-Time Effects Terminal

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Motivation and Background

Guitarists rely on multiple effects pedals to produce their own unique sound. While individual pedals can give guitarists a large assortment of effects, using multiple pedals can get messy quickly by increasing costs, requiring additional power and cables, and making things more difficult to change between effects. With the power of digital signal processing found in our project, many effects can be combined into a single programmable system, all inside of a guitar pedal interface.

A main drawback of these digital multi-effects pedals is that they crash and have poor tone at high volumes. Professionals steer clear because of this, despite their versatility and ability to produce an extremely wide array of sounds. This is due to overheating, high CPU utilization, and poor separation between analog signals and noise-producing components, and more. A better design for such kits would be to address one or more of these issues, improving the product by making it stage-ready, or closer to it.

Electric guitars produce an analog electrical signal that is modified by the user by implementing analog or digital effects pedals before being sent to an amplifier. Effects like distortion, delay, chorus, and reverb modify the contents of the guitar's analog electrical signal including amplitude, frequency, and phase. When using a digital effects system, the analog signal coming from the guitar must be converted into digital samples. These samples are then processed using algorithms before being converted back into an analog signal to be sent to an amplifier. This process is the fundamental design background for our project.

Our proposed system will seek to use a high-quality digital signal processor. The guitar signal will enter the pedal through a standard cable and pass through an analog input stage to condition the high impedance. An audio codec will perform the analog-to-digital conversion so that the digital signal processor can process the signal. Once the signal is processed, the audio codec will convert the digital signal back to analog. Finally, an output conditioning stage will ensure that the signal can be transferred to the amplifier. An additional microcontroller will also be used to manage most of the system's controls and peripherals, including potentiometers, encoders, and footswitches. A microcontroller is important to the project because it will allow the guitarist to interact with the pedal without needing a keyboard or mouse.

All these subsystems will be integrated through a custom PCB and will contain the microcontroller, power system, peripherals, and ADC/DAC. This will satisfy the design requirement of a significant custom PCB involving an MCU, power supply unit, and MCU peripherals. As per design instructions, pre-established reference circuits will be used for functions like voltage regulation, ADC/DAC, filters, etc.

The motivation for our project is to develop a digital multi-effects guitar pedal that will combine audio processing, physical control peripherals, and real-time audio-visual feedback into a single user-friendly device. It will receive an analog input from an electric guitar via a ¼" mono audio cable, which F.R.E.T will amplify then sample with an ADC for use with the digital signal processor. Once processed with the desired audio effects, the signal will be converted back to analog via a dedicated DAC and sent to a guitar amplifier via a ¼" mono audio cable.

This engineering project will be our team's opportunity to combine several areas of electrical and computer engineering that we have learned during our college careers into a single design.

Goals and Objectives

Goals

Basic Goals:

- Provide user with multiple guitar effects in a single device.
- Allow the user to control and adjust effects while playing the guitar.
- Allow the pedal to operate following an electric guitar and guitar amplifier.
- Provide reliable and safe operation from a single external power source.
- Create a self-contained and portable pedal suitable for normal guitar use.

Advanced Goals:

- Provide real-time audio visualizer that represents the guitar signal.
- Provide the user with clear information about the current effects and settings.
- Support simultaneous processing of ≥ 4 effects in a configurable signal chain, such as compression → distortion → chorus → reverb.
- Provide a streamlined user experience.

Stretch Goals:

- Allow the user to create and quickly recall customized sounds using presets.
- Allow the user to view how their enabled effects change the sound.
- Allow users to control and configure the pedal wirelessly.
- Provide additional useful tools to a guitarist, such as tuning and tempo control.
- Provide additional experimental guitar effects.
- Provide information about the operating condition of the pedal for troubleshooting.
- Allow the user to record data.
- Allow external devices to control the pedal.
- Improve system reliability through automatic fault detection and recovery.
- Allow the user to reprogram the MCU via USB or another external interface.

Objectives

Basic Objectives:

- Design and implement a real-time digital audio processing system capable of applying multiple guitar effects.
- Implement at least five digital guitar effects, including distortion, delay, chorus, compression, and reverb.
- Design an analog input and output signal path capable of interfacing the system with a standard electric guitar and guitar amplifier.
- Integrate an audio codec to perform ADC and DAC at ≥ 44.1 kHz, double the Nyquist frequency of the upper limit of human hearing.
- Achieve a minimum signal-to-noise ratio of ≥ 80 dB across intended audio bandwidth. This ratio is the standard
- Maintain an input-to-output audio processing latency of ≤ 10ms.

- Integrate physical footswitches and rotary controls that allow the user to enable effects and adjust effect parameters.
- Design a power subsystem capable of supplying all required system voltage rails from a single external DC power source.
- Design a single portable enclosure.

Advanced Objectives:

- Implement a graphical user interface that displays active effects, parameter values, and system information.
- Support the simultaneous processing of at least three digital guitar effects, such as distortion, compression, and reverb.
- Develop an effects-processing chain that allows individual effects to be enabled, disabled, and adjusted without interrupting the audio signal.
- Implement responsive interaction between the display, physical controls, and MCU.
- Maintain a user-interface response time of $\leq$ 100 ms for effect and parameter changes.

Stretch Objectives:

- Implement a preset system capable of storing and recalling at least 10 user-defined effect configurations.
- Achieve a signal-to-noise ratio of $\geq$ 90 dB
- Display the analyzed guitar signal as a digital spectrum visualizer with at least 16 frequency bands and a refresh rate of at least 20 updates per second.
- Implement before-and-after signal analysis to visually demonstrate how enabled effects modify the frequency content of the guitar signal.
- Implement Bluetooth communication for wireless configuration and control of the pedal.
- Implement a digital guitar tuner that identifies the detected musical note.
- Develop additional experimental digital effects such as bit crushing, reverse delay, auto-wah, or stutter effects.
- Monitor system operating information such as supply voltage, current consumption, processor temperature, or processing load.
- Implement system fault detection capable of identifying a loss of communication or processing failure and placing the pedal into a safe operating or bypass state.
- Implement a USB-compatible bootloader and accessible USB port.

Overall Marketing Goals:

Sound Quality, ensuring the user and audience can enjoy the sound without distraction.
Versatility, allowing the user to enjoy more features and sounds for less money and setup.
Stable Operation, allowing the end user to rely on our product for use in front of an audience and enjoy the experience our product can provide.
Cost effectiveness, providing the above goals for a reasonable price.
Size, allowing the end user to conveniently employ our product without impedance.

User-Friendliness, allowing our users to quickly and easily learn to use our product, with familiar controls and straightforward menus.
Component Quality, ensuring our users can enjoy a product with more robust, capable, and well-designed components.

Features/Functionalities

- **Real-Time Processing.** Digitalized guitar signal for audio processing and converted back to analog for output with low latency. No delay between playing a note and hearing it.
- **Versatile Effects Library.** The most common effects such as distortion, delay, chorus, compression, and reverb, are all included in one single pedal.
- **Tactile Control Surface.** Dedicated footswitches for hands-free effect switching, eliminating mid-song breaks to achieve desired effects. Rotary encoders provide effect customizations, such as changing the amount of distortion, tone shaping, and controlling volume.
- **Instant Presets.** Store and recall user-created effect configurations, including enabled/disabled state and parameter values via physical controls.
- **Simultaneous Effects.** User-selectable combination of active effects allowing multiple to run at once.
- **Optimal Display.** User interface is shown on a compact LCD screen without compromising readability.
- **Simple I/O.** A single guitar input and single amplifier output, using standard ¼" jacks.
- **Status Indicators.** Footswitches paired with LEDs to show active effects.

Existing Related Projects

Pi-Stomp V3

As shown in Figure 1(a), the Pi-Stomp V3 is an open-source, DIY multi-effects pedal kit built around a Raspberry Pi 5. It runs a full Linux pro-audio stack, mod-host, and MOD-UI, hosting hundreds of LV2 effects and amp-modeling plugins, including Neural Amp Modeler support. Physical controls consist of four footswitches and a set of rotary encoders, paired with a small onboard color display for navigating presets. This pedal is popular among DIY pedal enthusiasts due to community documentation and support. Users can install community-made effects or write their own. It sells as a $299 kit and represents the "affordable, hackable" end of the multi-effects spectrum. Despite being made by a small company, Tree Fall Sound, the Pi-Stomp has amazing sound quality. The pedal includes a codec connected to the Raspberry Pi, which supports a sampling rate of 96kHz. Known issues regarding the pedal include slow boot time, software delays at high frequency, and overheating, which can cause the system to crash. [A.1]

Sonicake Matribox

As shown in Figure 1(b), the Sonicake Matribox is a budget-tier, closed-hardware multi-effects processor, with a lineup spanning around $150-350 depending on tier (base Matribox, Matribox II, and the touchscreen Matribox II Pro). The current Matribox II ($229.99) offers 63 amp models, 93 effect/pedal simulations, and 61 cabinet simulations,

running up to 10 simultaneous, reorderable effect modules on a color LCD screen. On the lower tier, processing is capped at 24 -bit/44.1kHz, which is lower than competitors who aim for 96kHz. Provided with only three footswitches, and no XLR output, the lower tier models can't offer users a steady gigging experience. The same core signal-processing remains in higher tier models such as the Matribox II Pro. The differences include additional footswitches, XLR outputs, and a full color touchscreen. [A.2]

FunBox

As shown in Figure 1(c), FunBox is an open-source, nine single-effect guitar pedal platform, built around the Daisy Seed chip by Electro-Smith. The MCU on the Daisy Seed is the ARM Cortex-M7 (STM32H750), running at 480MHz. The board is equipped with stereo audio IQ, 24-bit/96kHZ audio hardware. This is more focused on the DIY aspect of pedal building. For the price range, the FunBox comes in at the lowest, which is under $100. There are flaws in the system such as having no signal chain (each individual effect pedal needs a physical cable to connect), re-flashing firmware required to switch effect, no bypass relay, and no display screen. [A.3]



(a) (b) (c)

Figure 1. Existing projects: (a) Pi-Stomp V3, (b) Sonicake Matribox, and (c) Funbox.

Engineering Specifications

Engineering specifications refer to the quantitative measurement and requirements for both the system as a whole (Table 1), and the individual components contained within (Table 2). These values are determined by assessing the marketing and engineering goals, then determining what quantitative measurements may achieve those goals.

Overall System Engineering Specifications			
#	**Engineering Specification**	**Quantitative Measure**	**Verification Method**
1	Audio sampling rate	≥ 44.1 kHz	Verify the codec and software sampling configuration
2	Audio processing latency	≤10 ms guitar input ->processed audio output	Apply input waveform and measure the time difference between input and output with oscilloscope

3	Startup time	≤30s from power-on to operational audio processing	Measure from power application to operation
4	User interface response time	≤100 ms from user input to displayed parameter change	Measure the time between control inputs and screen response
5	UI refresh rate	≥ 20FPS	Measure the update period
6	Audio frequency range	20 Hz – 22.05 kHz through audio signal path	Frequency sweep through the pedal and measure the output
7	Continuous operation	≥ 1 hour without crashing or audio loss	Operate the system continuously for one hour
8	Effects library	≥ 5 selectable digital effects	Demonstrate five different effects and the use of preset feature
9	Simultaneous digital effects	≥ 3 effects operating simultaneously	Enable maximum effects and visually and audibly confirm
10	Signal-to-noise ratio	≥ 80 dB	Apply input waveform and measure the output voltage and output noise for calculation

Table 1. System Engineering Specifications Table

Component Engineering Specifications	
Custom PCB	
PCB type	Multilayer PCB
Integration	MCU, power circuitry, audio interfaces, and peripheral interfaces
Fabrication features	Solder mask and silkscreen
MCU	
CPU Cores	2 cores
Maximum Primary CPU Clock	480 MHz
On-Chip Flash Memory	2 MB
On-Chip RAM	≥ 1 MB
Floating-Point Processing	Hardware FPU
Audio Communication Interface	≥ 1 I²S/SAI interface

Peripheral Communication	SPI, I²C, UART, GPIO
Supply Voltage	Between 1.7 V and 3.6 V
I/O Pins	≥156 pins
Signal Processing Capacity	≥ 1000 DMIPS
Audio Codec	
ADC/DAC Resolution	≥ 24 bits
Audio Sampling Rate	≥ 48 kHz
Number of Audio Channels	≥ 2 channels
ADC Dynamic Range	≥ 90 dB
DAC Dynamic Range	≥ 90 dB
Digital Audio Interface	I²S compatible
Analog Supply Voltage	3.3 V or 5 V compatible
Display	
Display Size	4.0–4.3 in diagonal
Display Resolution	≥ 800 × 480 pixels
Display Type	IPS TFT LCD
Color Depth	≥ 16-bit color
Refresh Rate	≥ 20 frames/s
Display Communication Interface	RGB / parallel interface
Operating Voltage	3.3 V or 5 V compatible
Rotary Encoders	
Type	Incremental rotary
Output	Quadrature incremental digital output
Range	360-degree rotation
MCU Interface	GPIO compatible
Included hardware	Encoders and knobs
Footswitches & Buttons	
Type	Momentary
MCU interface	Digital GPIO
Included hardware	Footswitches, buttons, and mounting hardware
Audio Input Stage	
Connector	1/4-in (6.35 mm) TS
Input type	Mono guitar input

Input conditioning	High-impedance buffer
Frequency range	20 Hz–20 kHz
Protection/filtering	Input protection, filtering, and AC coupling
Audio Output Stage	
Connector	1/4-in (6.35 mm) TS
Output type	Mono audio output
Output impedance	≤ 1 kΩ
Output conditioning	Buffer, filtering, protection, and coupling
Power	
External Input Voltage	9–12 V DC
Digital Supply Rail	3.3 V ± 5%
Auxiliary Supply Rail	5.0 V ± 5%
3.3 V Rail Current Capacity	≥ 1.5 A
5 V Rail Current Capacity	≥ 1.0 A
Regulator Efficiency	≥ 85% at nominal load
Output Voltage Ripple	≤ 50 mV peak-to-peak
Load Capacity	≥ 125% of calculated maximum system load
Input Protection	Reverse-polarity and overcurrent protection
External Power Adapter	
Output	Regulated DC
Output voltage	9–12 V DC
Current capacity	≥100% of maximum system input-current requirement
Connector	Compatible with pedal DC input
Status Indicators	
Indicator type	LEDs
Function	System/effect status indication
Colors	Red/Green/Blue

Table 2. Component Engineering Specifications Table

House of Quality

The House of Quality is a visual representation of the various goals of the product, the specifications that comprise those goals, and the tradeoffs between pursuing each goal. The marketing goals can be seen on the left, denoting what our product prioritizes in a broad

sense. In the figure below (Figure 2), the highest priority is written in the 1) slot, second highest in 2), and so on.

The vertical section represents quantifiable specifications useful for determining how well the product's marketing priorities are being fulfilled. A "+" sign means that we seek to maximize the associated value, and conversely, a "-" sign means we seek the opposite. An up arrow means that the intersection between the two characteristics is positively correlated, e.g. maximizing sound quality and minimizing latency do not oppose one another. Furthermore, two upwards arrows indicate a strong positive correlation. Similarly, a single downwards arrow means a negative correlation, and two downwards arrows mean a strong negative correlation, e.g. minimizing CPU Temperature and maximizing Effects Count strongly oppose one another.

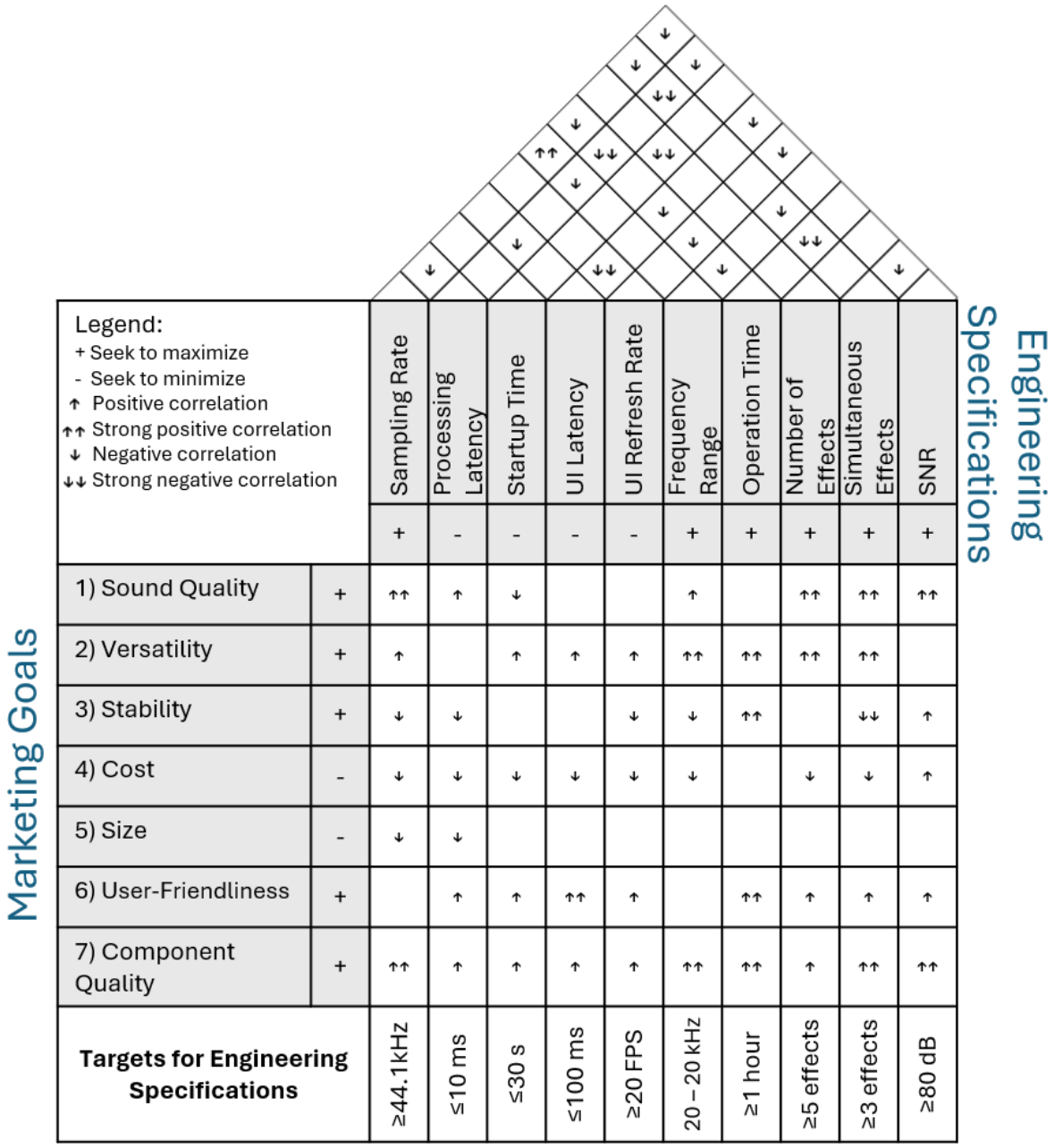


Figure 2. House of Quality

Hardware Block Diagram

A hardware block diagram is a useful overview of a system, providing a visual representation of major design choices as well as the interaction between components. In the below hardware block diagram (Figure 3), we seek to represent the multi-effects pedal as a general approach to solving the problem and achieving the marketing goals.

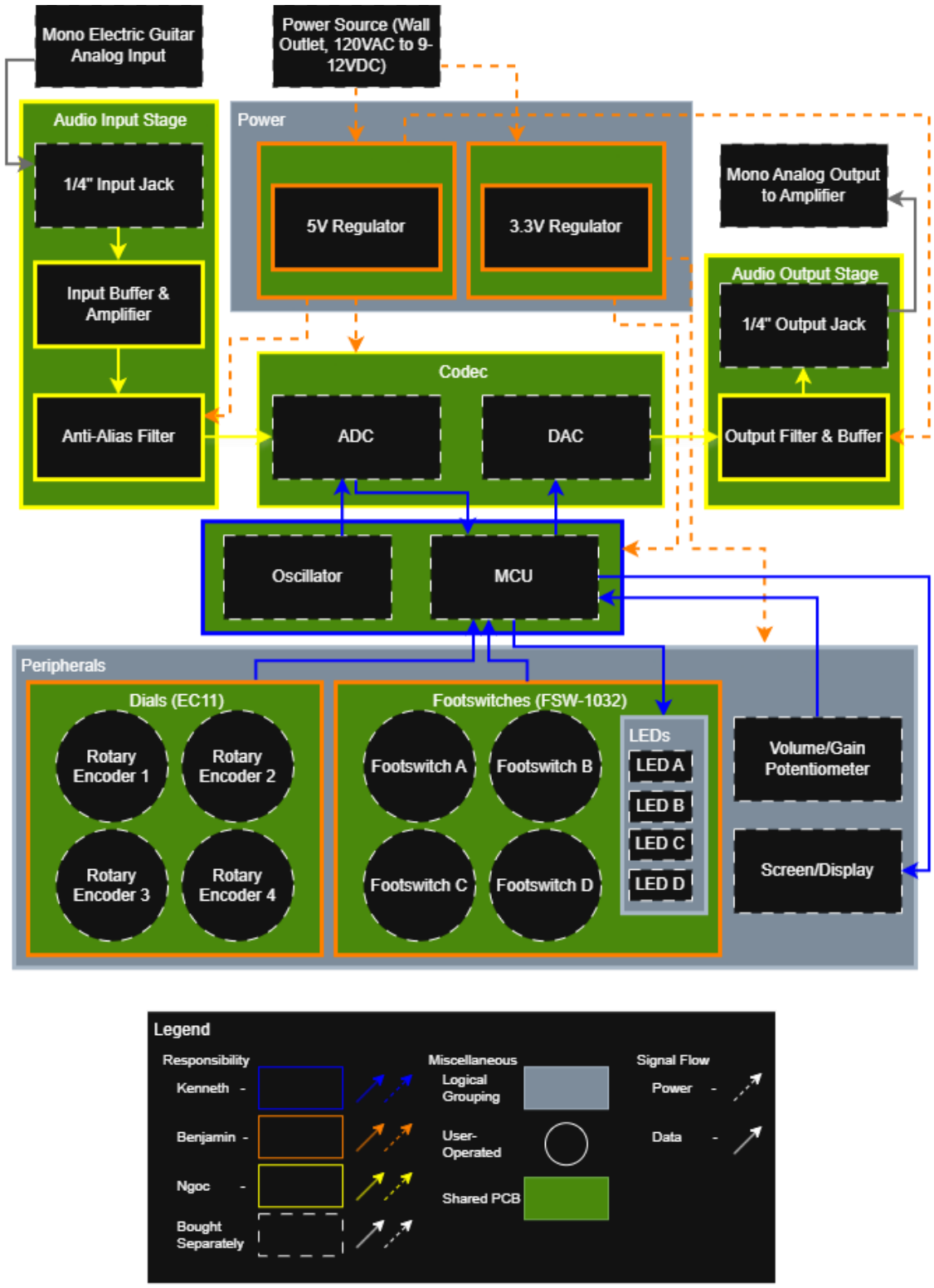


Figure 3. F.R.E.T. Hardware Block Diagram

To do this, we need to support a DSP device and allow the user to interact with the system. The user will be able to interact with input device(s), allowing control of the system, while the embedded hardware is supported by an input stage, output stage, and power subsystem.

Software Flowchart of Overall System

The user will utilize rotary encoders, buttons, footswitches, and a display to operate F.R.E.T. To accomplish this, a software controller will be created or customized based on existing controllers, if suitable candidates can be found and appropriate permissions from the creators obtained. The controller will allow the peripherals to act as an interface between the user and the custom effects. Operation of the user interface is logically shown via flowchart in Figure 4.

The current design proposal involves utilizing a number of rotary encoders and buttons to operate the UI. The primary rotary encoder can be rotated to move the selection indicator between neighboring software elements (namely, the Effects tab and the Presets tab, or Effect 1, 2, 3, etc.). The user can press the button within the primary rotary encoder to interact with the selected tab, activating further UI depending on the selected tab.

In the above mock-ups, the selected effect is surrounded in one color, while other tabs are indicated in another. Furthermore, an effect written in green is active (Intensity $\geq 0\%$), while an effect written in red is inactive (Intensity = 0). Intensity is an effect parameter that can be altered by selecting the specific effect the user wishes to change.

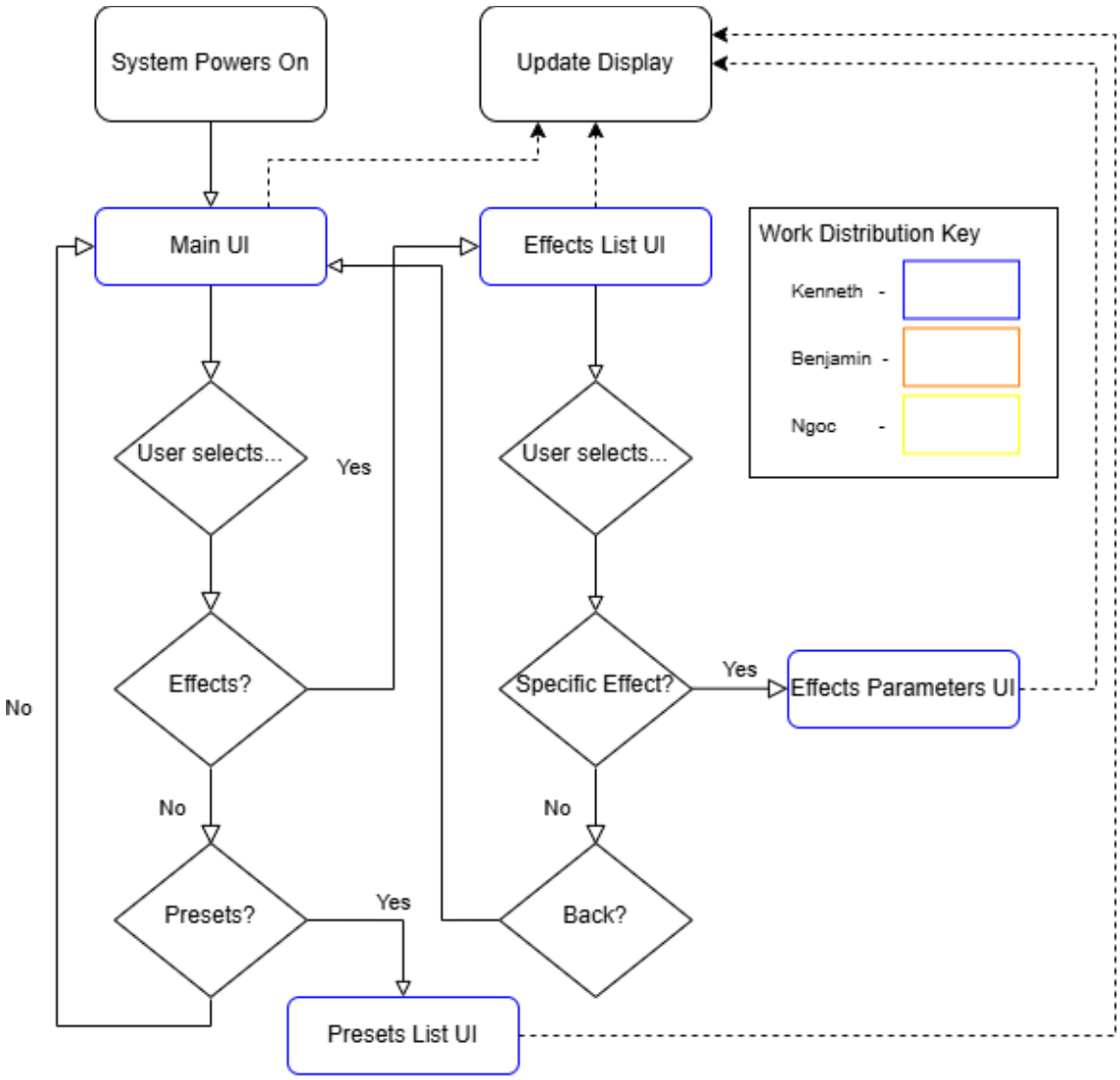


Figure 4. F.R.E.T. UI Software Block Diagram

Figure 4 shows how initial user interaction with F.R.E.T should be responded to. This diagram is not an exhaustive list of all UI elements, since further interactable menus involve much more processing, requiring a more complex and extensive diagram. More detail is required for both the “Presets List” UI, and the “Effects Parameters” UI. The software diagram for the Effects Parameters UI can be found in Figure 5, while the software diagram for the Presets UI can be found in Figure 6.

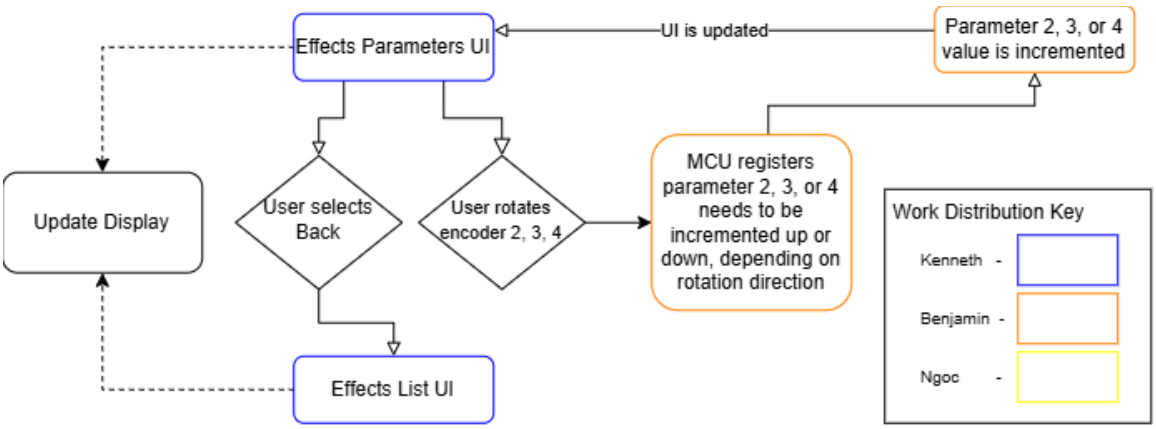


Figure 5. Effects Parameters UI, Software Block Diagram

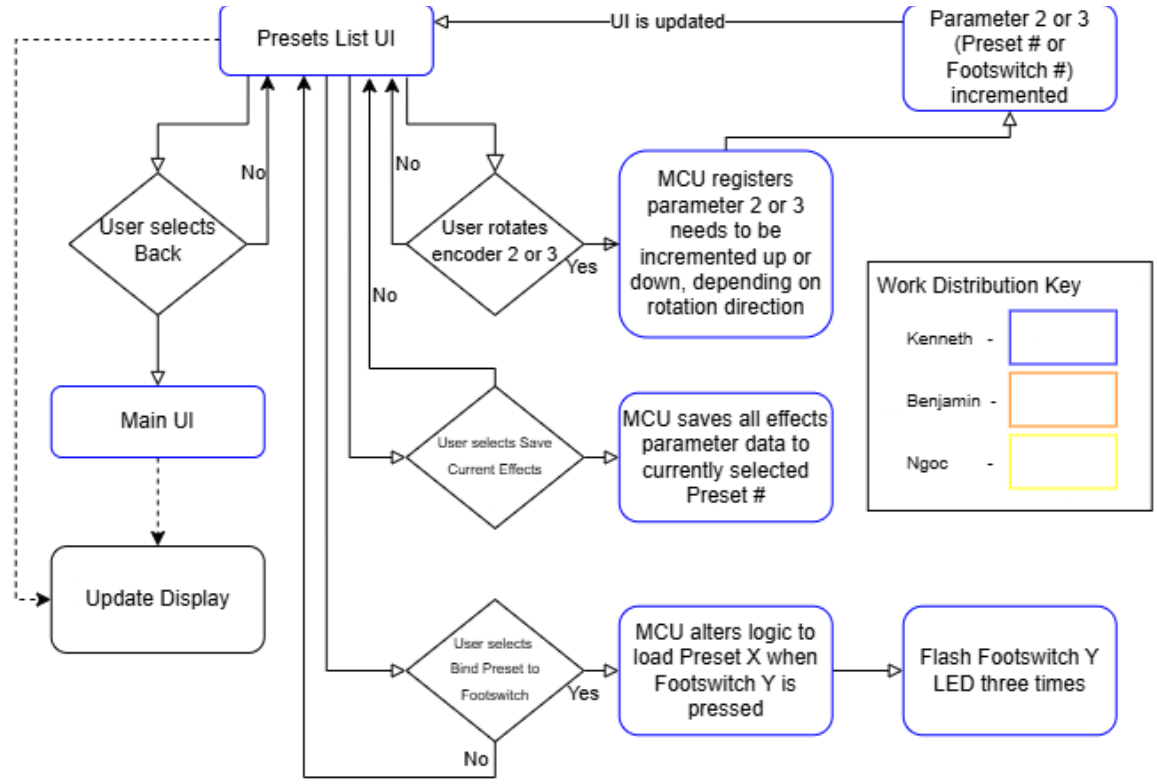


Figure 6. Presets UI, Software Block Diagram

Via the above mechanism, the user can save their effects to a preset. To access these saved presets, they can use the Presets List menu to bind a preset to a footswitch, allowing them to load the preset on demand by pressing the footswitch.

Additionally, if F.R.E.T detects audio input and simultaneously has not detected any input from the user via the peripherals (rotary encoders, footswitches, or buttons) for an interval

of time, it will automatically display the audio visualizer. A software block diagram for this feature can be seen in Figure 7.

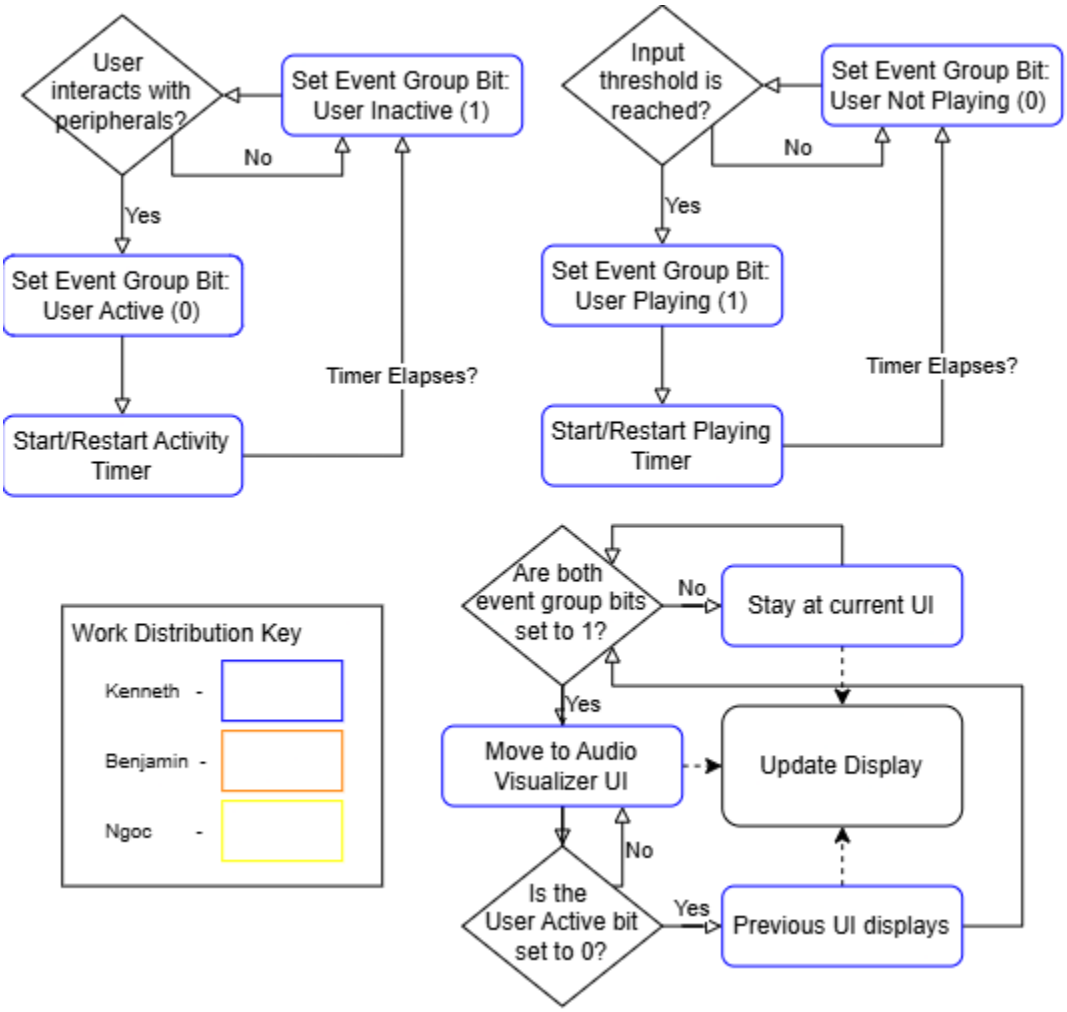


Figure 7. Audio Visualizer Software Block Diagram

Importantly, the digital signal processor must also apply the selected effects to the incoming audio signal, in addition to providing the UI. While the effects themselves may perform simple or complex signal processing, the underlying mechanism is straightforward, and can be seen in the following diagram (Figure 8):

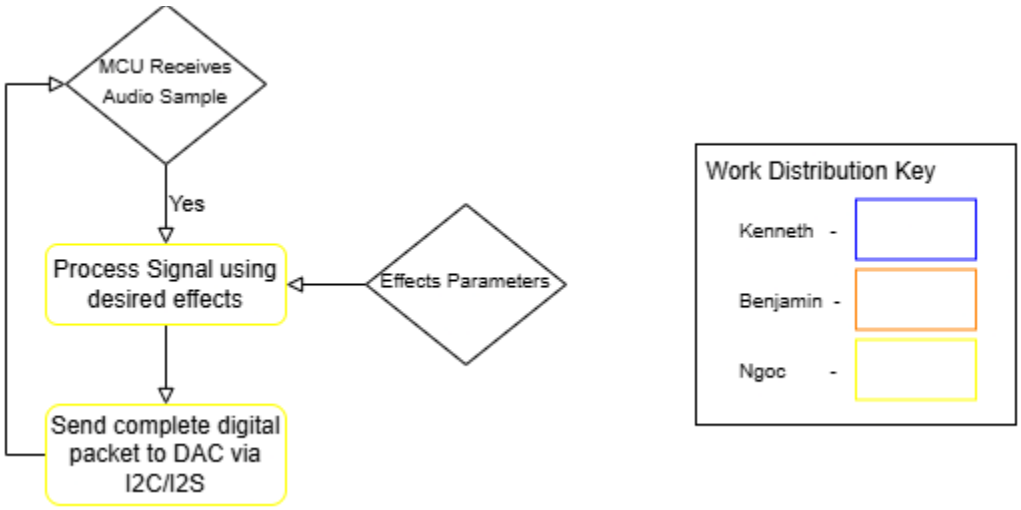


Figure 8. Digital Processing Software Block Diagram

Prototype Illustration/Blueprint

The blueprint (Figure 9) is built based on the design of the Pi-Stomp V3, with slight dimension changes to accommodate the larger display. The blueprint shows our general idea of what F.R.E.T should look like; this will most likely change in the future. One thing to note is that the spacing of the footswitches will stay relatively the same. They are 2.75 inches apart, for users to switch between effects without accidentally selecting a different open. Figure 10 shows the normal use case of F.R.E.T.

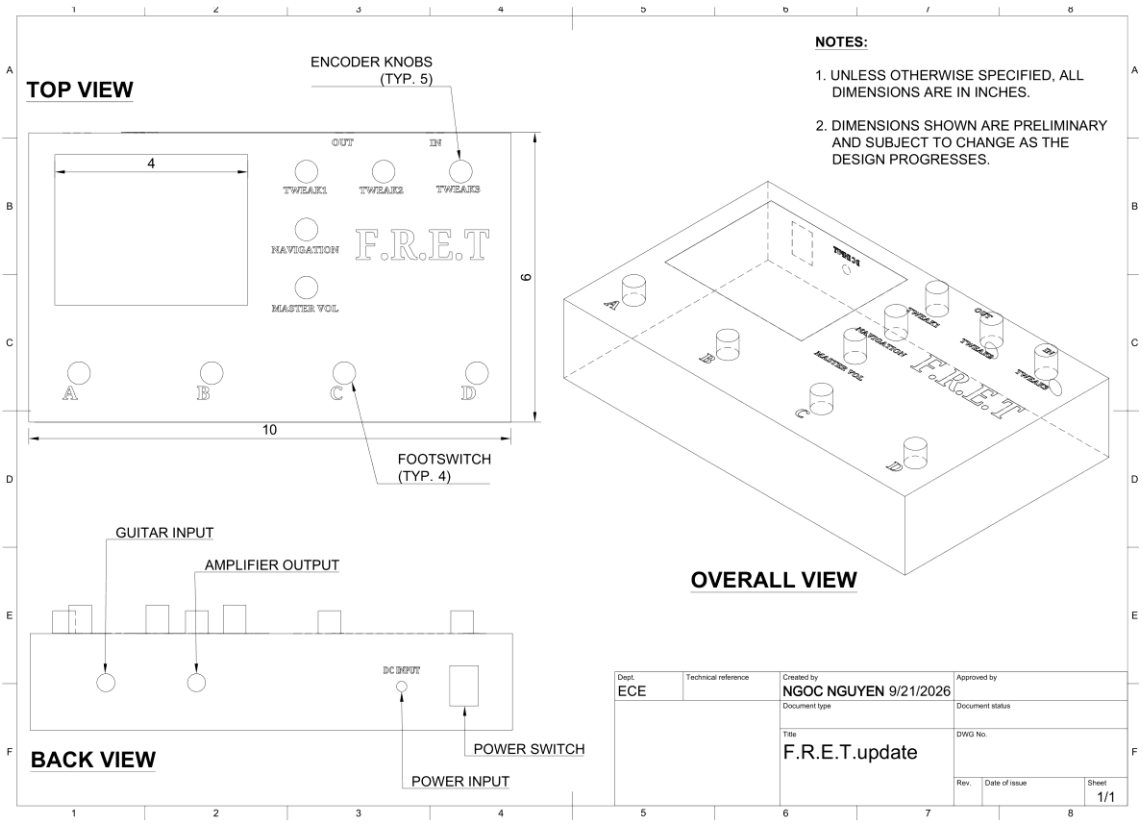


Figure 9. F.R.E.T. Tentative Blueprint

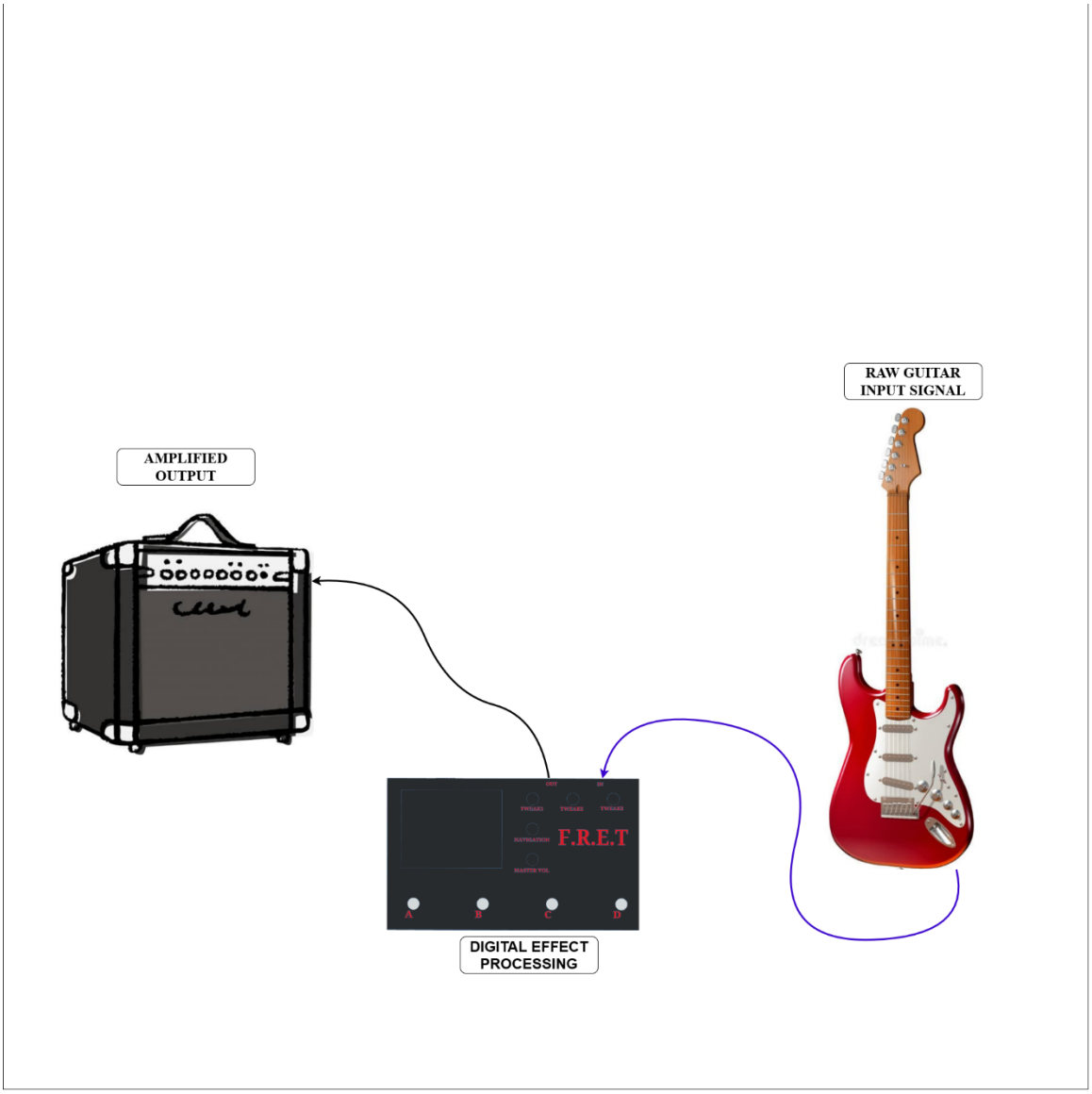


Figure 10. F.R.E.T. Illustration

Based on the illustration, the pedal stands between the guitar and the amplifier. Even though the guitar and amplifier are not in scope, they will be included to demonstrate the functions of the pedal.

Budget and Financing

The existing products mentioned are supplement guides for our project. The target is to build a hybrid product using good qualities of the existing products, while being able to improve certain aspects. We want to build a beginner friendly, affordable product without sacrificing quality. See Table 3 for an estimated budget.

Hardware	Description / Included Components	Estimated Budget

Custom PCBs	PCB fabrication, multilayer board, solder mask, silkscreen, and shipping	**$80**
MCU Dev Board	Development board for prototyping; final MCU incorporated into custom PCB	**$30**
Audio Codec	ADC/DAC audio codec and supporting interface components	**$20**
Display	TFT LCD display	**$40**
Rotary Encoders	Rotary encoders and knobs for effect and parameter adjustment	**$10**
Footswitches & Buttons	Footswitches, push buttons, and associated mounting hardware	**$20**
Audio Input Components	1/4-inch input jack, high-impedance buffer, filtering, protection, and coupling components	**$10**
Audio Output Components	1/4-inch output jack, output buffer, filtering, protection, and coupling components	**$10**
Power Input & Protection	DC connector, fuse/PTC, reverse-polarity protection, TVS protection, and input filtering	**$10**
Voltage Regulators	Switching regulators/LDOs and supporting inductors, capacitors, and resistors	**$10**
External Power Adapter	DC power adapter capable of supplying the complete pedal	**$10**
Status Indicators	LEDs and current-limiting resistors for system and effect status	**$5**
PCB Connectors & Headers	Display connector, programming/debug header, internal connectors, and pin headers	**$10**

Enclosure	Pedal enclosure for the complete system	**$20**
Enclosure Hardware	Standoffs, screws, nuts, washers, feet, and PCB/display mounting hardware	**$5**
General Electronic Components	Resistors, capacitors, diodes, transistors/MOSFETs, op-amps, and miscellaneous passives	**$20**
Wiring & Assembly Supplies	Hookup wire, ribbon cable, heat shrink, solder, and miscellaneous assembly supplies	**$10**
Contingency / Replacement Parts	Spare components, replacements, and unexpected hardware expenses	**$10**
	Total	**$330**

Table 3. Project Budget

Project Milestones for SD1 and SD2

To remain on track, and provide structure to a long-term goal, we've come up with a tentative schedule of tasks which will help to facilitate timely completion of important milestones (Table 4).

Task	Dates	Responsibility
Develop project proposal	9/4-9/25	Benjamin, Ngoc, Kenneth
Update Committee members	9/25-10/2	Benjamin, Ngoc, Kenneth
Prepare and design website for continued use	9/25-10/2	Kenneth
Initial parts acquisition	9/25-10/9	Benjamin, Ngoc, Kenneth
Design input and output buffer circuits	9/25-10/16	Benjamin
Begin writing embedded software	10/2-10/16	Kenneth, Ngoc
Design dev board MCU circuitry (peripherals, I/O)	10/2-10/16	Kenneth
Design power supply circuits	10/2-10/16	Ngoc
Update Committee and Website	10/9-10/16	Benjamin, Ngoc, Kenneth
Prototype and troubleshoot designed circuits in isolation. Compare strengths and weaknesses.	10/9-10/16	Benjamin, Ngoc, Kenneth
Implement prototyped circuits integrated with one another. Compare strengths and weaknesses.	10/16-10/23	Benjamin, Ngoc, Kenneth

Revise prototyped circuits, determine best implementation/design	10/16-10/23	Benjamin, Ngoc, Kenneth
Revise embedded software	10/16-10/23	Kenneth, Ngoc
Design standalone MCU board (move away from dev board)	10/16-10/23	Kenneth
Update Committee and Website	10/16-10/23	Benjamin, Ngoc, Kenneth
Write Midterm Report	10/16-11/6	Benjamin, Ngoc, Kenneth
Design and order buffer, power, MCU PCBs with components	10/23-11/6	Benjamin, Ngoc, Kenneth
Revise Midterm Report	11/6-11/13	Benjamin, Ngoc, Kenneth
Assemble, test, and troubleshoot PCBs	11/10-11/18	Benjamin, Ngoc, Kenneth
Update Website and Committee	11/13-11/20	Benjamin, Ngoc, Kenneth
Begin filming Mini Demo video	11/13-12/3	Benjamin, Ngoc, Kenneth
Begin Final Report	11/13-12/3	Benjamin, Ngoc, Kenneth
Revise PCB design and order updated version	11/13-11/18	Benjamin, Ngoc, Kenneth
Assemble PCBs, ver. 2	12/3-1/8	Benjamin, Ngoc, Kenneth
Senior Design 2 Begins	1/11	Benjamin, Ngoc, Kenneth
Update Website and Committee	1/11-1/15	Benjamin, Ngoc, Kenneth
Implement and Test Design	1/15-1/29	Benjamin, Ngoc, Kenneth
Major Revisions, or Begin Advanced and Stretch Goal designs	1/29-2/19	Benjamin, Ngoc, Kenneth
Update Website and Committee	1/29-2/5	Benjamin, Ngoc, Kenneth
Parts Acquisition	2/5-2/19	Benjamin, Ngoc, Kenneth
Update Website and Committee	2/19-2/26	Ngoc
Reattempt Basic goals, or Begin Advanced and Stretch Goal Prototypes	2/5-2/26	Benjamin, Ngoc, Kenneth
Design and order PCBs	2/26-3/5	Ngoc
Refine and polish design	3/19-4/16	Benjamin, Ngoc, Kenneth

Live Demo and Final Review Meeting	4/23-5/5	Benjamin, Ngoc, Kenneth

Table 4. Senior Design Milestones

Work Distribution

In order to facilitate the timely completion of F.R.E.T, objectives have been approximately divided into different sections and assigned to each member of the group, paying attention to previous experience, skill level, desire to learn a skill, and workload fairness. See Table 5 below for specifics.

Overall, Kenneth has interest and experience with real-time embedded systems (both in bare-metal programming as well as RTOS' including FreeRTOS), hardware assembly, PCB design, GitHub, HTML/CSS, and CAD software. In addition to primarily focusing on MCU hardware, software, assembly/testing, and communications, he also has an interest in contributing to PCB design, modeling the enclosure, and assisting with the development of input/output circuitry.

Ngoc has experience building filters and amplifier circuits. She is most interested in designing the PCB layout, audio input/output circuitry, and power supply design. She is also interested in applying CAD software to design the enclosure for the project. Additionally, she would like to contribute to the software development of the system to gain further experience with embedded software and digital signal processing.

Benjamin will be responsible for designing the power supply unit (PSU) and protective circuitry to provide stable power and protect the system from potential electrical faults. He will also handle the risk management by identifying possible hardware failures before they occur and find methods to improve the system's safety and reliability. He is interested in this area because it will give him hands-on experience with power electronics and circuit protection while maintaining the system's health.

Task	**Primary member**	**Secondary member(s)**
PCB Design	Ngoc	Kenneth and Benjamin
Audio Input/Output Design	Ngoc	Kenneth
CAD Software	Ngoc	Kenneth
PSU Design	Benjamin	Ngoc
Protective Circuitry, Risk Management	Benjamin	Kenneth
Scheduling, Deadlines and Accountability	Benjamin	Kenneth and Ngoc
MCU/Peripheral Design	Kenneth	Benjamin
UI, DSP, Visualizer Software	Kenneth	Ngoc
Hardware Assembly/Testing	Kenneth	Benjamin
Website/Communication with Committee	Kenneth	Benjamin

Table 5. Work Distribution

Appendix A – Reference

[A.1] Tree Fall Sound. “Pi-Stomp Manual.” *Pi-Stomp Manual - Pi-Stomp Manual*, treefallsound.github.io/pistomp-manual/home/. Accessed 11 Sept. 2026.
[A.2] Sonicake. “SONICAKE QME-50 Matribox Multi-Effects Processor.” *Sonicake*, www.sonicake.com/products/matribox. Accessed 11 Sept. 2026.
[A.3] Bloemer, Keith. “GuitarML/Funbox: Stereo Guitar Pedal Platform Using Daisy Seed.” *GitHub*, github.com/GuitarML/FunBox. Accessed 11 Sept. 2026.

Appendix B – Copyright and Usage Permission

Currently, no outside intellectual properties have been used in the creation of this document.

Appendix C – Miscellaneous